Filipe Xavier de Oliveira

engfilipeoliveira89@gmail.com | ntknowledge.com
GitHub: parapapinho | LinkedIn: Filipe Oliveira

ABOUT

I am a security researcher focused on Windows internals, reverse engineering, and vulnerability research. My professional background includes Windows kernel vulnerability research, malware analysis, penetration testing, and security assessments. I have many years of experience developing software in assembly language with MASM (Microsoft Macro Assembler).

My research interests also include processor architecture and microarchitecture, instruction set architectures (ISAs), processor vulnerabilities, and transient execution attacks. I enjoy debugging and reverse-engineering anti-cheat systems used in games to understand their internals and protection mechanisms.

Beyond my professional roles, I have been involved in the reverse engineering community as a member of CrackLatinos, where I have been active since 2004. I specialize in analyzing and unpacking complex software protections, including Enigma, Armadillo, Themida, and VMProtect. I also enjoy participating in reverse engineering CTFs and challenges such as Flare-On.

I share technical knowledge through articles and presentations on C and assembly programming, vulnerability analysis, and kernel mitigations.

SKILLS AND ABILITIES

Reverse Engineering and Vulnerability Research

- Static and dynamic analysis of executables, libraries, and Windows kernel drivers.
- User-mode and kernel-mode debugging with IDA and WinDbg to investigate code execution, memory, and system behavior.
- Extensive experience analyzing and unpacking software protected with Enigma, Armadillo, Themida, and VMProtect.
- Analysis of anti-debugging techniques, code obfuscation, and software protection mechanisms.
- Crash dump analysis, crash triage, and root cause investigation.
- Deep understanding of Windows internals, including the interaction between user-mode applications and kernel components.
- Vulnerability research involving memory corruption, information disclosure, and local privilege escalation.
- Exploit development and proof-of-concept implementation to validate vulnerabilities and assess their impact.
- Malware research through reverse engineering and behavioral analysis of malicious artifacts.
- Computer architecture and x86/x64 assembly, including calling conventions, stack organization, and memory access.
- Assembly programming with MASM, plus C and PowerShell, including library development for RadASM and Easy Code and scripting for security analysis.
- Debugging and reverse engineering of game anti-cheat systems.

Security Consulting and Penetration Testing

- Extensive experience conducting security assessments and penetration tests across web applications, APIs, mobile applications, infrastructure, core systems, and IoT devices.
- Specialized security assessments of ATM and POS environments for Brazilian financial institutions.
- Manual testing of authentication, authorization, session management, and business logic.
- Source code review to identify security flaws, investigate their root causes, and support remediation.
- Adversary simulation and APT emulation to assess security controls and evaluate SOC detection and incident response capabilities.
- Development of proofs of concept and attack scenarios to demonstrate exploitability, chain related weaknesses, and assess business impact.
- Technical reporting and communication of findings to engineering teams, security teams, and stakeholders, with practical remediation guidance.
- Remediation validation and retesting to verify the effectiveness of security fixes and identify incomplete corrections.

EXPERIENCE

Tempest Security Intelligence | Jan 2016 - Present
Principal Security Analyst

- Conduct end-to-end security assessments across web and mobile applications, infrastructure, and mission-critical systems.
- Perform malware research and reverse engineering within threat intelligence operations, analyzing malicious code and attacker behavior to support security teams.
- Support incident response by investigating malicious activity and technical evidence to determine scope and impact and guide containment and remediation.
- Design and execute adversary simulations for SOC teams, emulating real-world attacker tactics, techniques, and procedures to evaluate detection coverage and incident response capabilities.
- Lead advanced penetration testing and secure code review engagements, including critical ATM and POS security assessments for Brazilian financial institutions.
- Deliver cryptography-focused security assessments for major enterprise clients, evaluating cryptographic implementations and identifying weaknesses in their design and application.
- Research vulnerabilities in publicly available software, coordinate responsible disclosure, and reverse-engineer proprietary client libraries to uncover complex security flaws.
- Develop and deliver internal technical training on computer architecture, reverse engineering, and exploit development.

Allele Security Intelligence | Oct 2024 - Jan 2026
Senior Vulnerability Researcher

- Conducted Windows kernel vulnerability research through reverse engineering and analysis of kernel components.
- Discovered multiple privilege escalation vulnerabilities in the Windows kernel.

ESET Latin America Research Laboratory | 2009 - 2011
Malware Researcher

- Conducted technical malware research and targeted threat intelligence analysis of malicious campaigns.
- Authored internal white papers and technical briefs analyzing newly observed malware techniques.

EDUCATION

Degree in Software Systems Development | 2018 - 2021
Faculdades Metropolitanas Unidas (FMU), Brazil

CERTIFICATES AND TRAINING

- Corelan - Expert-Level Stack-Based Exploit Development | Nov 2024
- Allele Security Intelligence - Linux Userland Exploit Development | Sep 2022
- Allele Security Intelligence - Linux Kernel Exploit Development | Mar 2022

PUBLIC VULNERABILITY RESEARCH

- Windows Kernel Information Disclosure - CVE-2026-32215, CVE-2026-32217, CVE-2026-32218
- G DATA Total Security 25.4.0.3 ActiveX Buffer Overflow - CVE-2018-10018
- Panda Global Protection 17.0.1 Insecure Named Pipe Permissions - CVE-2018-6322
- Bitdefender Total Security 2018 Insecure Named Pipe Permissions - CVE-2018-6183
- SyncBreeze 10.1.16 XML Parsing Stack-Based Buffer Overflow - CVE-2017-15950
- Handy Password 4.9.3 Buffer Overflow - CVE-2017-17946

TECHNICAL WRITINGS

- What Every Hacker Should Know About Floating-Point Numbers | Slides
- A Study on C Integers | Slides
- User Stack: Essential Knowledge for Studying Memory Corruption

ONLINE PUBLICATIONS

- Unpacking Enigma Protector 7.40 Like the Old Days: Bypassing Anti-Debugging, Finding the OEP, and Rebuilding the IAT
- The Breakpoint I Never Set: How IDA's F8 Triggered an Anti-Debug Check
- Analyzing a Kernel Address Disclosure via Windows Event Log — CVE-2026-32217
- Exploring the Windows Kernel | Video | Slides

LANGUAGES

Portuguese (native); Spanish (fluent); English (advanced comprehension; conversational speaking).